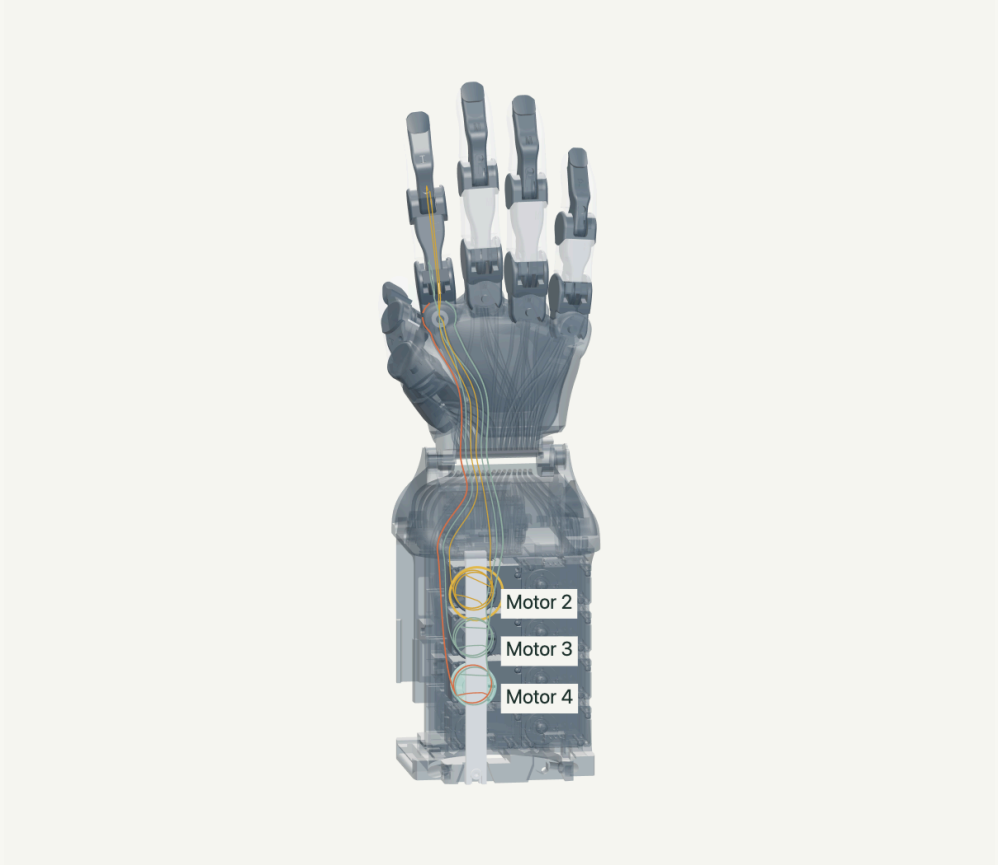
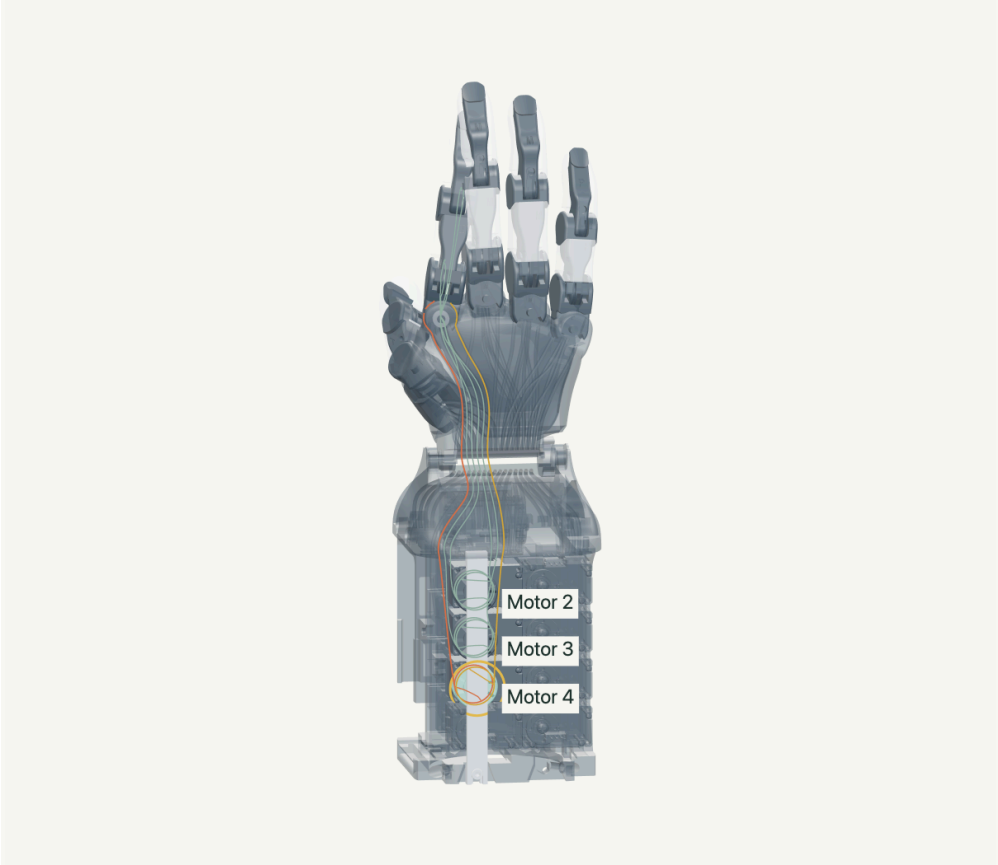
01 Identify the affected motor

A finger has six tendon ends; the thumb has eight. Keep the other ends attached to their spools. Pop out only the parts needed for access. Keep their pins and bearings fitted.



Wiggle the finger; watch the matching motor.

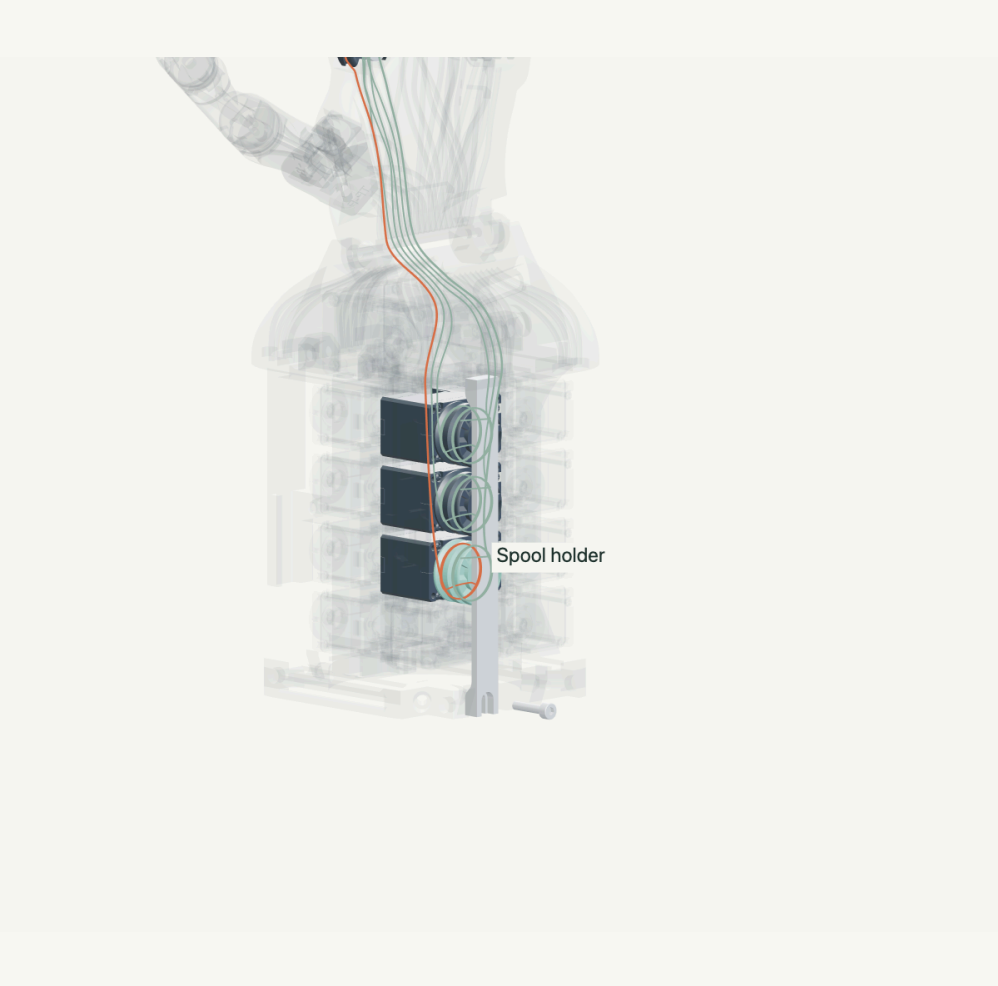


Trace the broken AP tendon to motor 4.

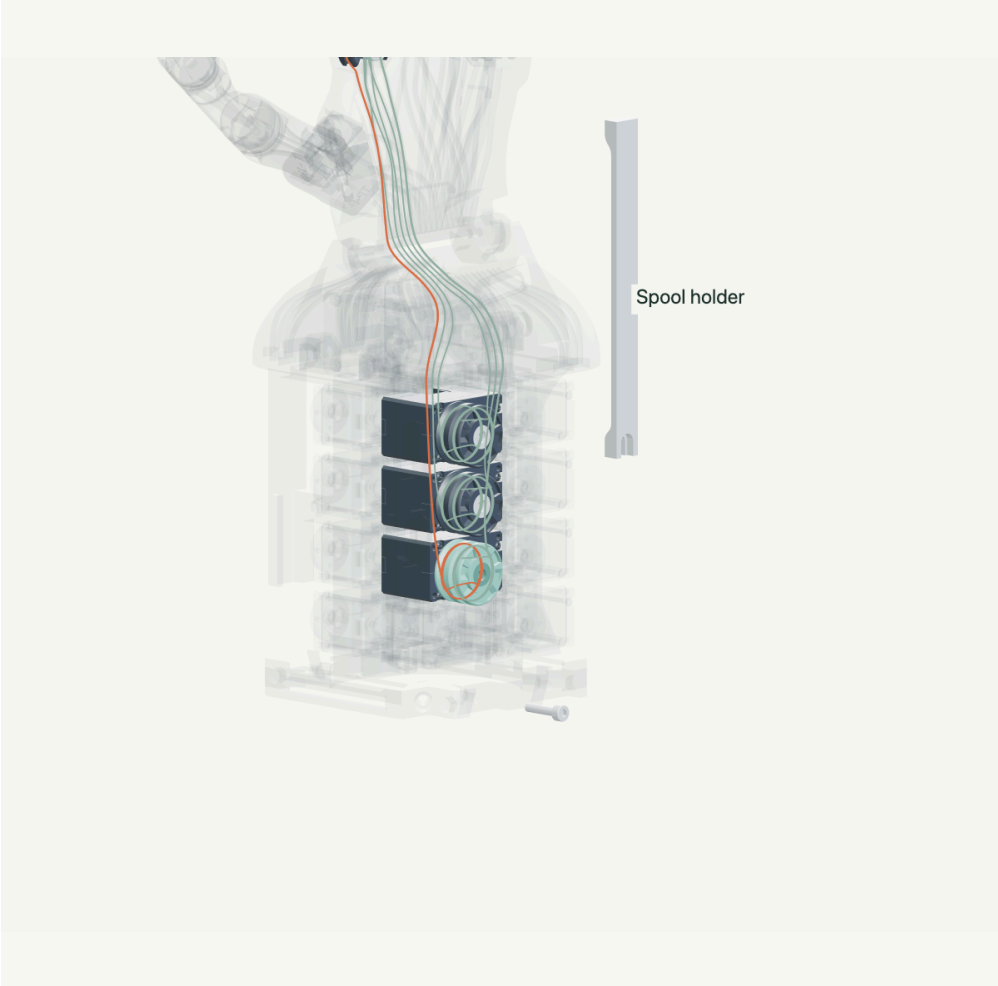
Disconnect and power off. Gently wiggle the finger to identify its motors. Trace the broken tendon to its motor; record each cord’s spool and winding direction.

Stop the operation, Disconnect and switch board power off first. A broken cord may not transmit movement: trace its opposing cord and confirm the motor with the matching assembly guide. Keep the recorded top/bottom assignments.

02 Remove the spool holder



Remove the holder screw.

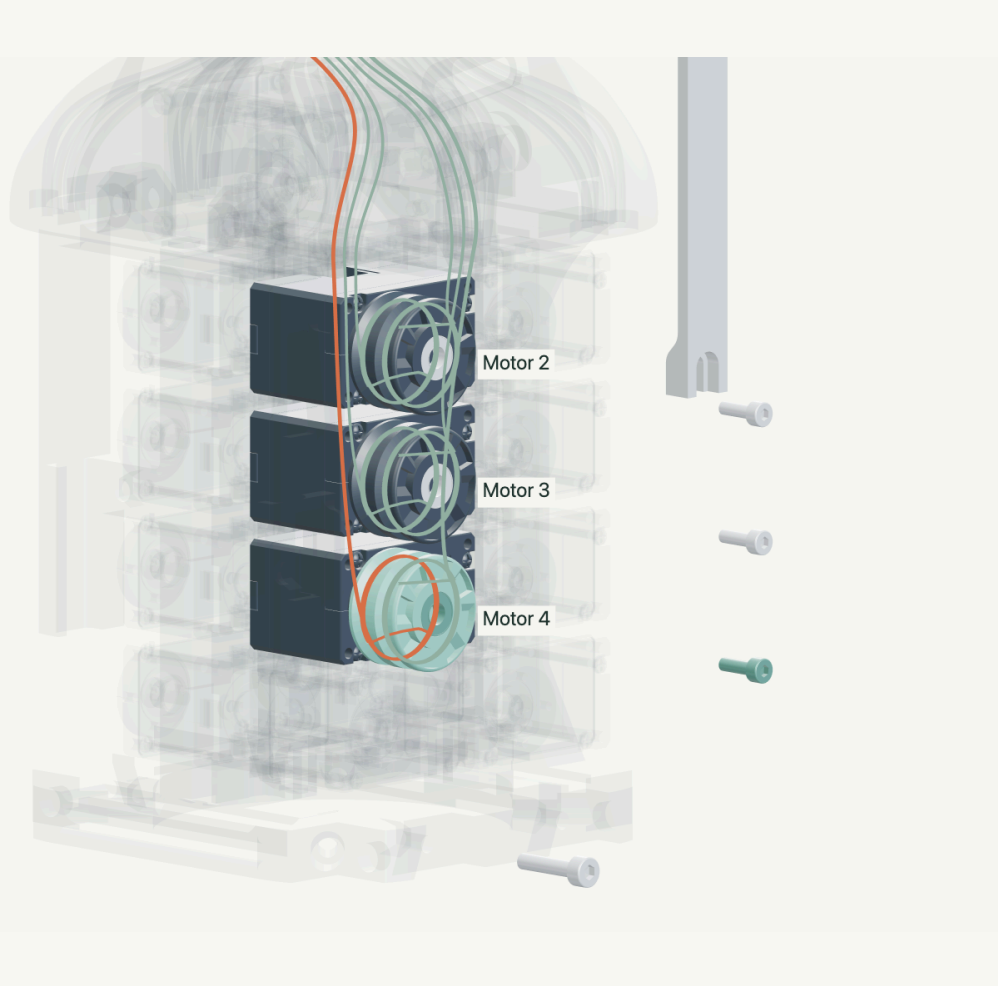


Slide the holder clear of the spools.

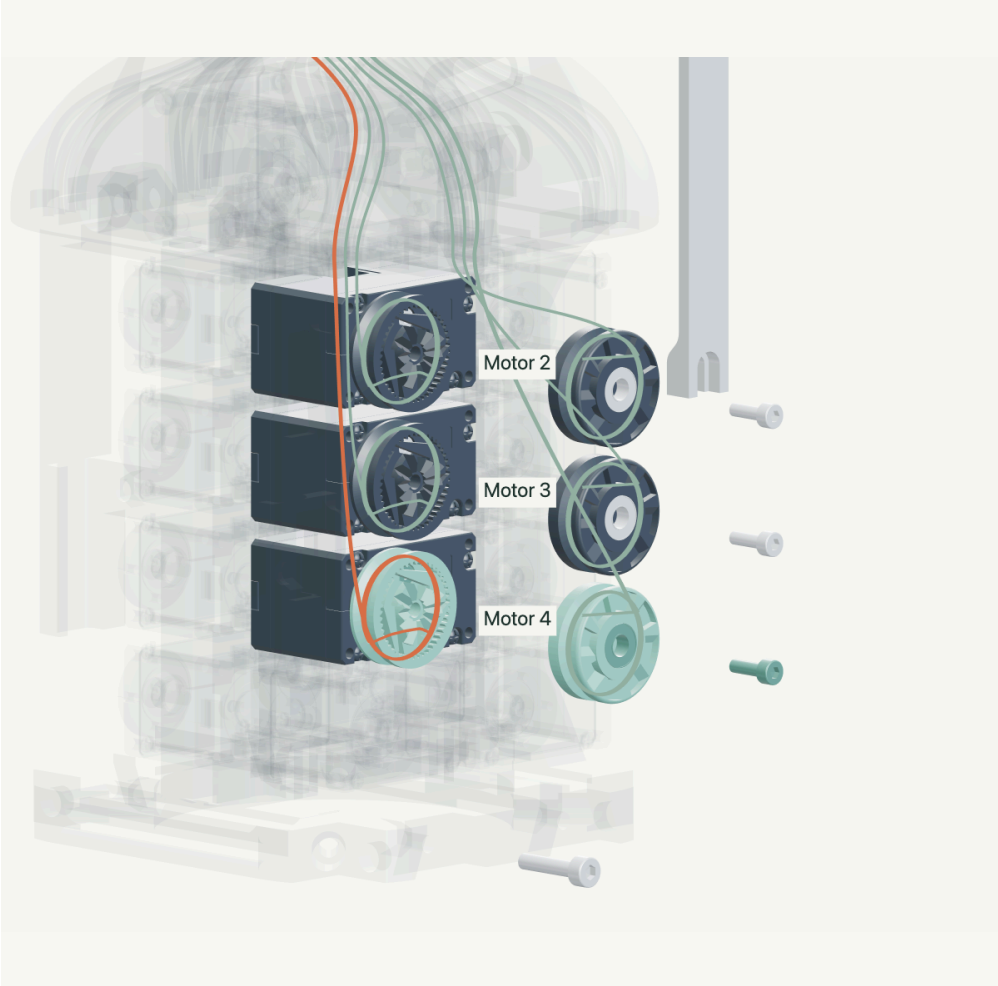
Undo the holder’s screw. Slide the holder out of its slots and put it aside.

Keep the holder and its matching screw together, clear of the motor column.

03 Lift the top spools



Remove the top-spool screws.

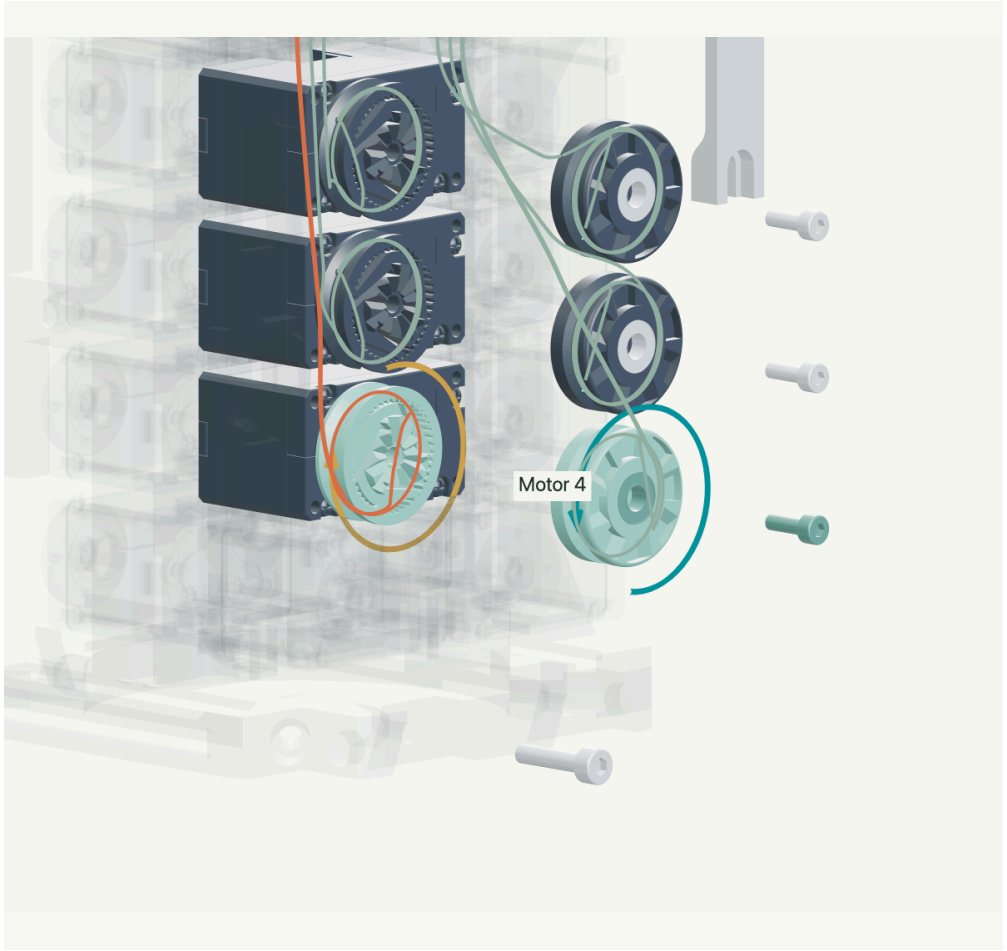


Lift the top spools; keep their bearings fitted.

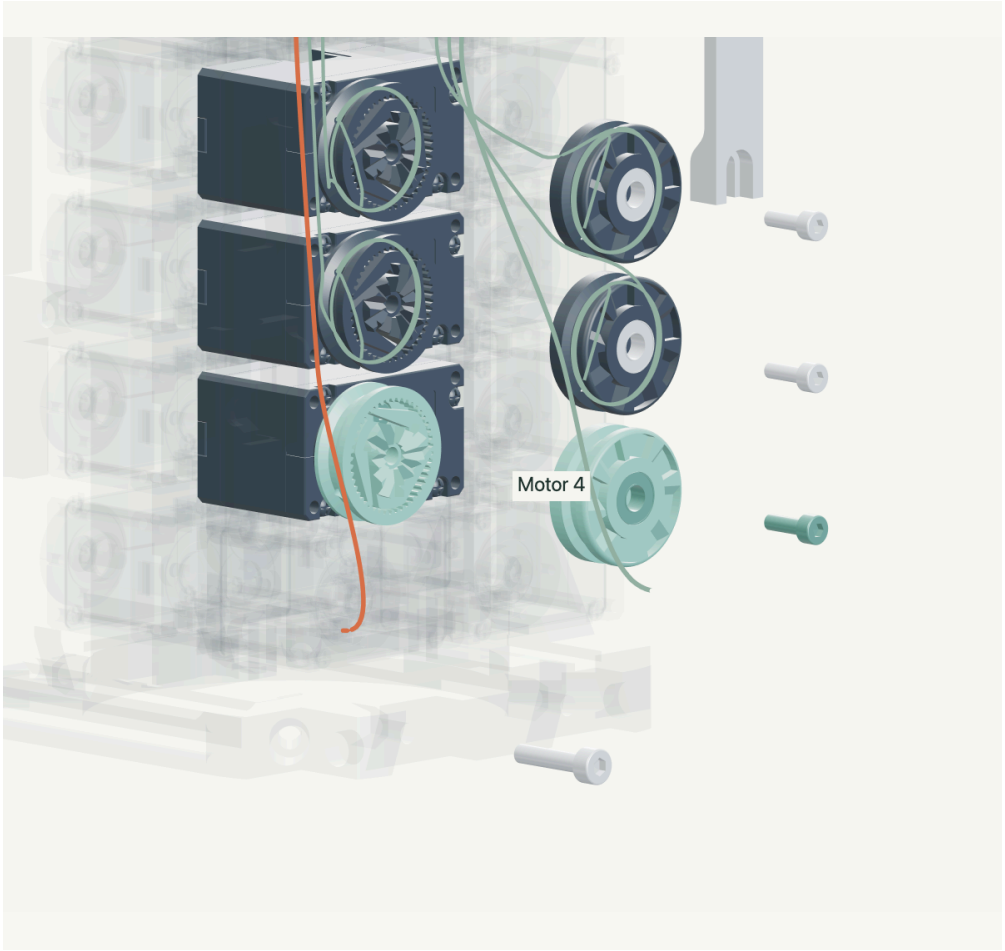
Remove the top-spool screws for this finger. Lift off the top spools with their bearings still seated.

Keep each top spool with its original cord. Leave the bottom spools on the motors. Do not pull bearings out of the top spools.

04 Unwind the tendons



Counter-rotate to unwind the motor 4 pair.

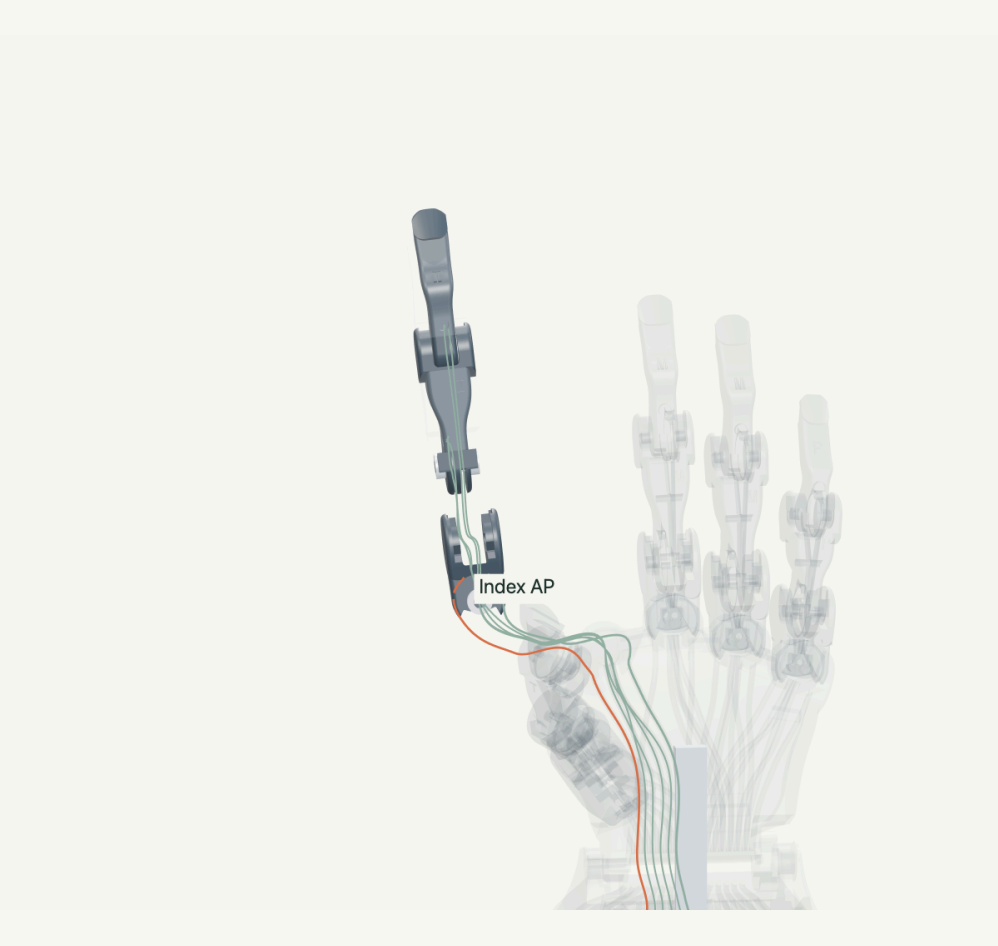


Free both ends from the spool channels.

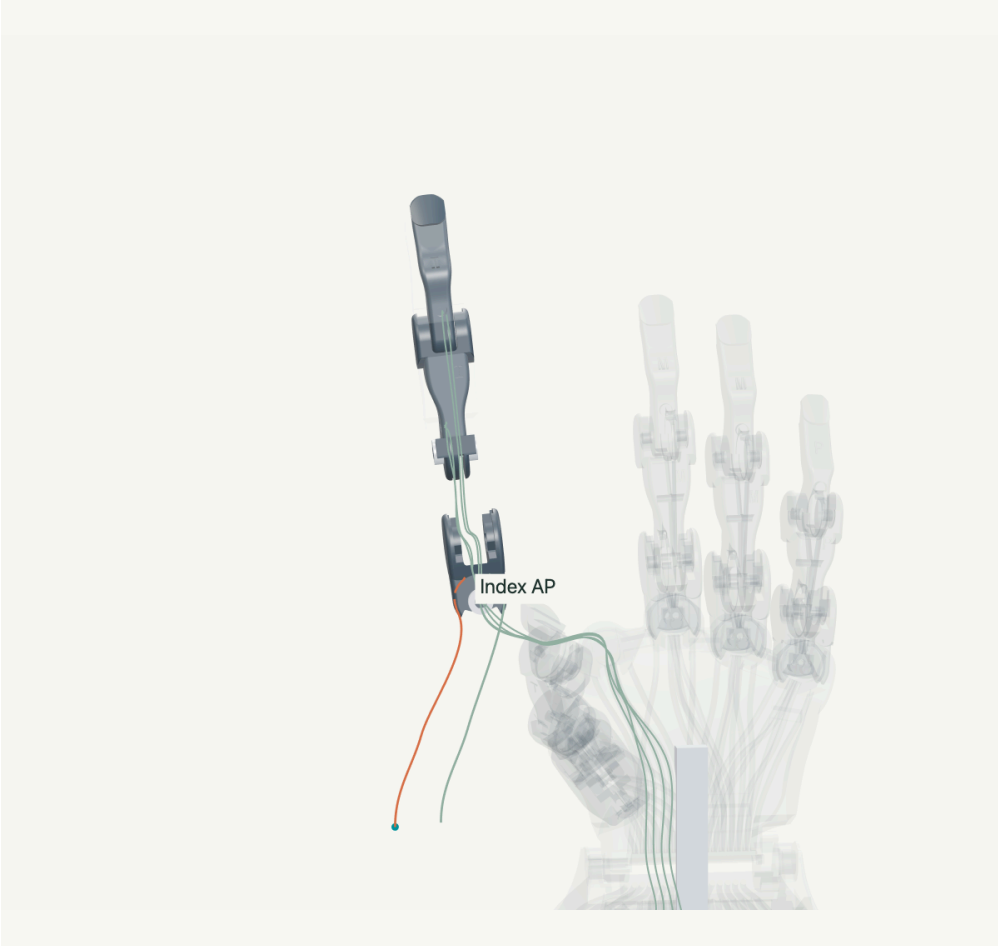
Counter-rotate the spools to slacken all six ends. Fully unwind the motor 4 pair and free both ends from their spool channels before opening the finger. Keep the other four ends anchored.

Follow your recorded winding direction in reverse. Motors 2 and 3 keep their four cord ends anchored. Only the motor 4 pair comes fully off its spools.

05 Pop the finger out



Pop out the finger; lift off the tip-and-middle unit.

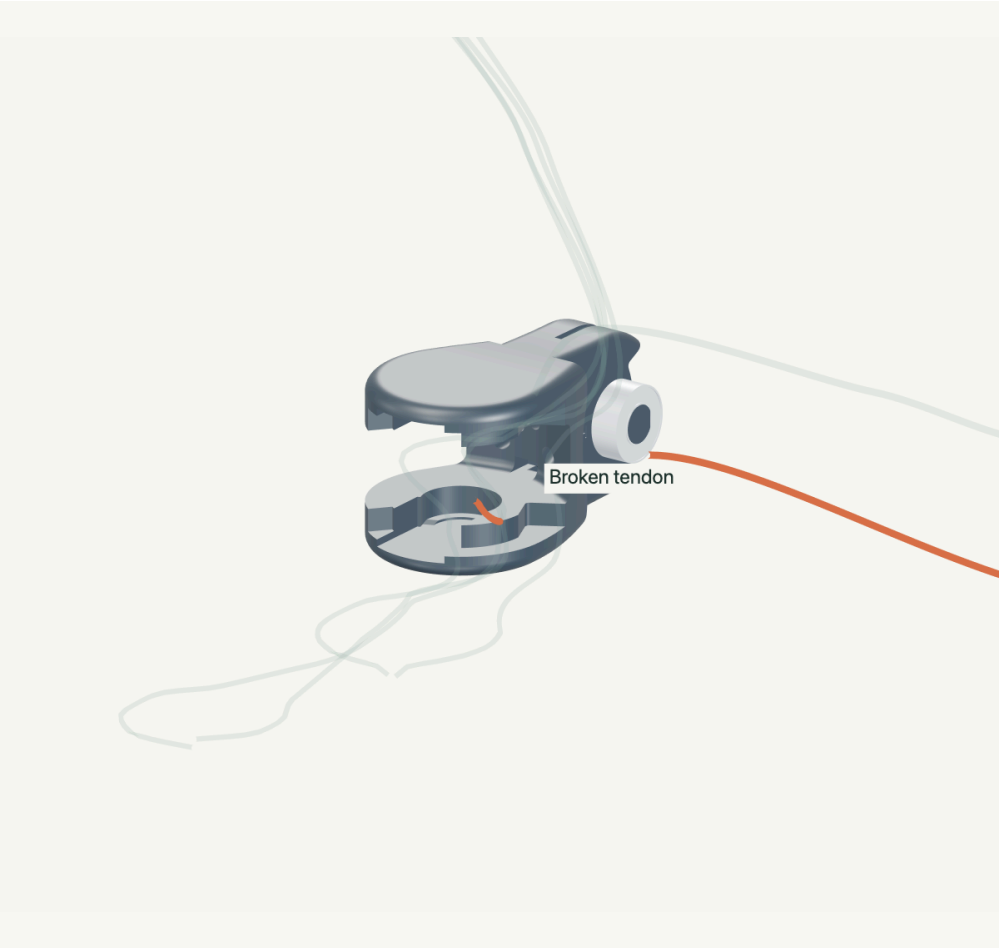


Draw the AP pair back through the wrist and carpals.

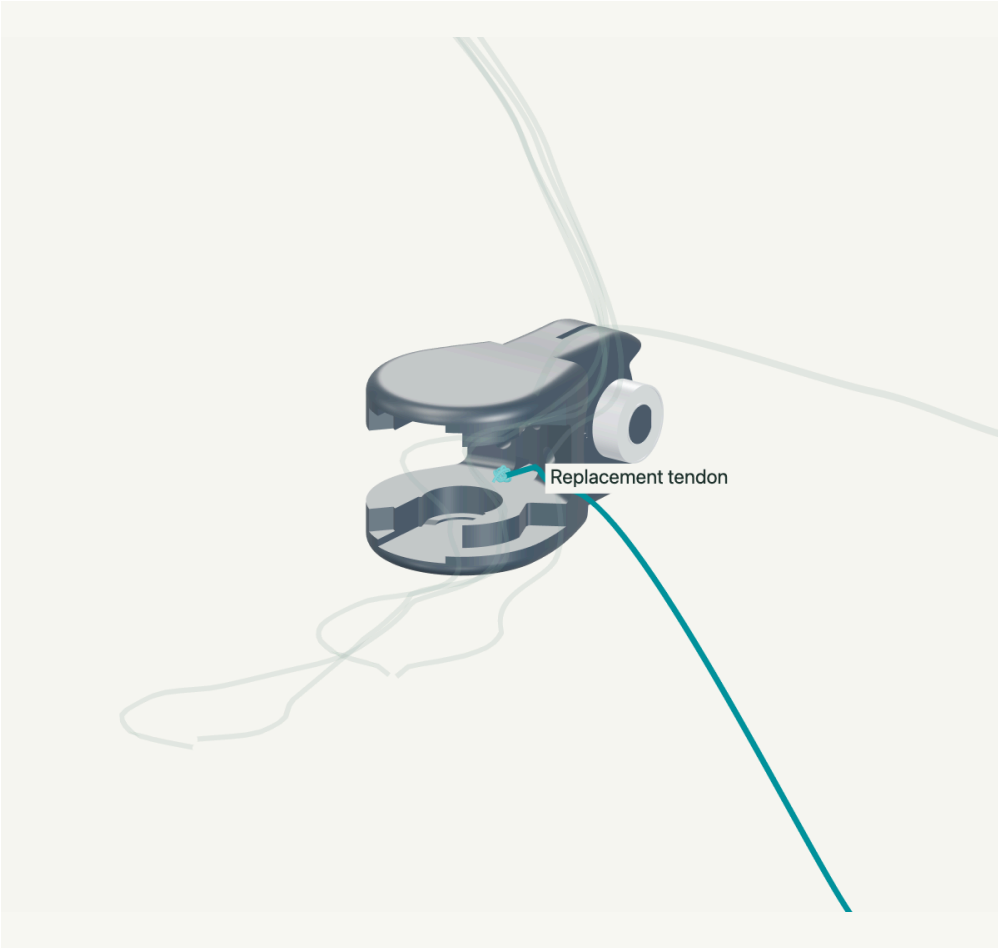
Pop out the index finger, then lift its tip-and-middle unit off the AP. Keep all pins and bearings fitted. Pull the two freed AP ends back through the wrist and carpals.

Keep the pins and bearings in their parts. Leave intact cords attached to their spools and move them only as far as access requires. Do not drag an anchored knot through a channel.

06 Replace and tie the cord



Pull out both pieces of the broken cord.



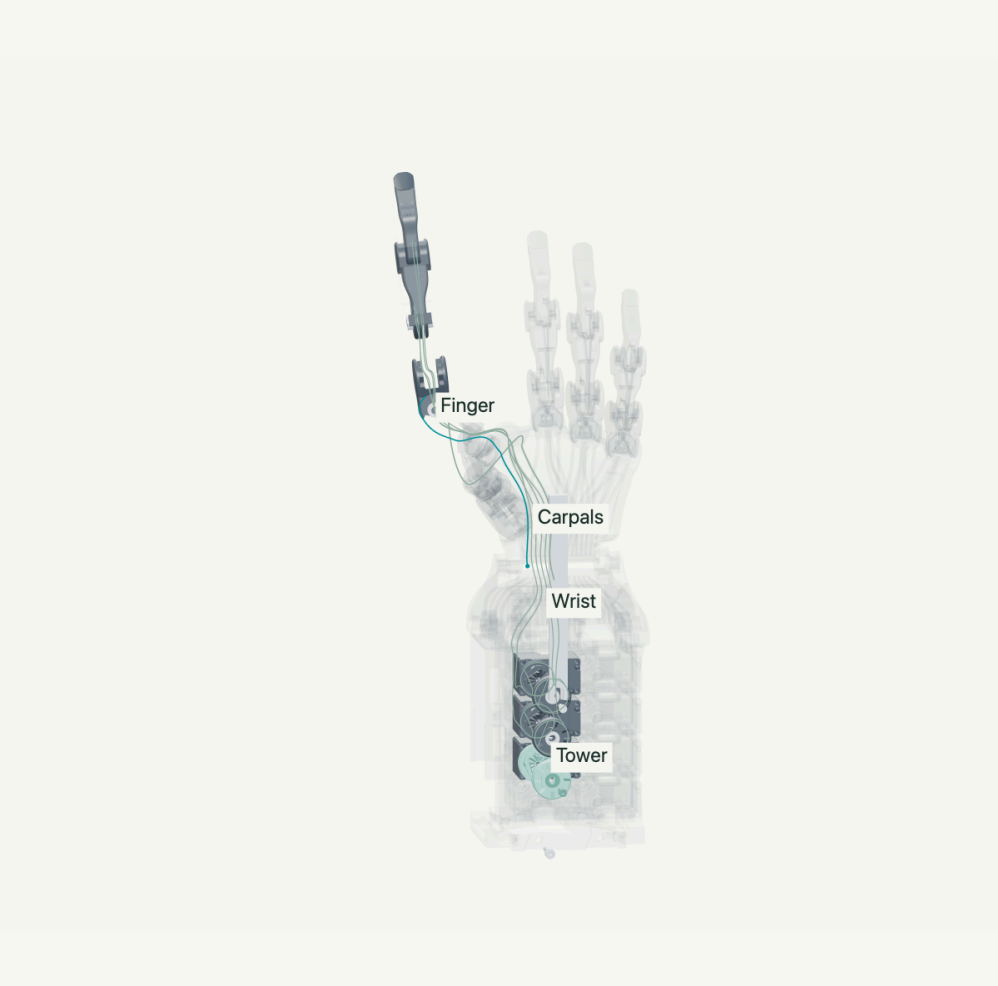
Insert the replacement, tie the stopper and seat it.

Pull out the damaged cord and broken pieces. Insert the new cord. Tie and seat the same anchor knot used in the assembly guide.

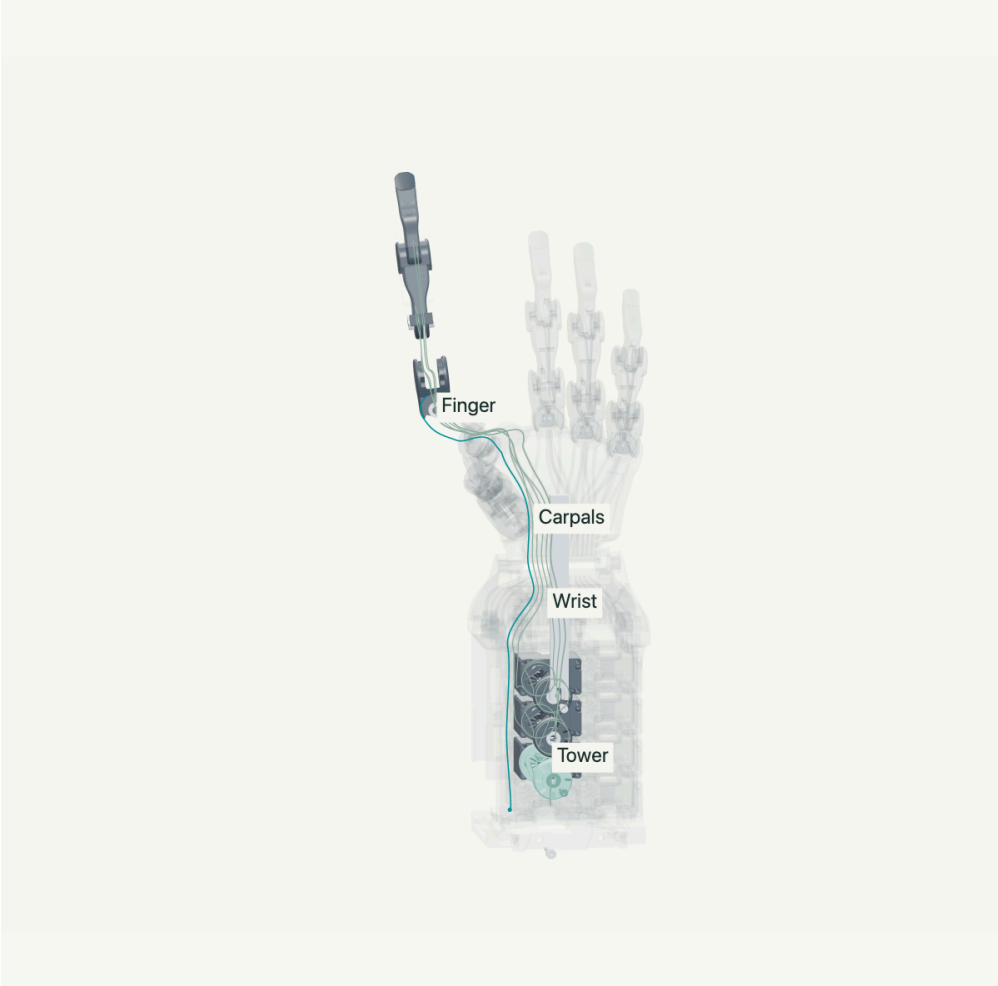
Check the liner before inserting new cord. Tie an Ashley stopper, then one single knot directly over it. Seat the knot in the AP anchor. Reuse the opposing AP cord if undamaged.

Ashley stopper: illustrated knot sequence on pages 10–11.

07 Thread back to the tower



Follow the measured carpal channels.

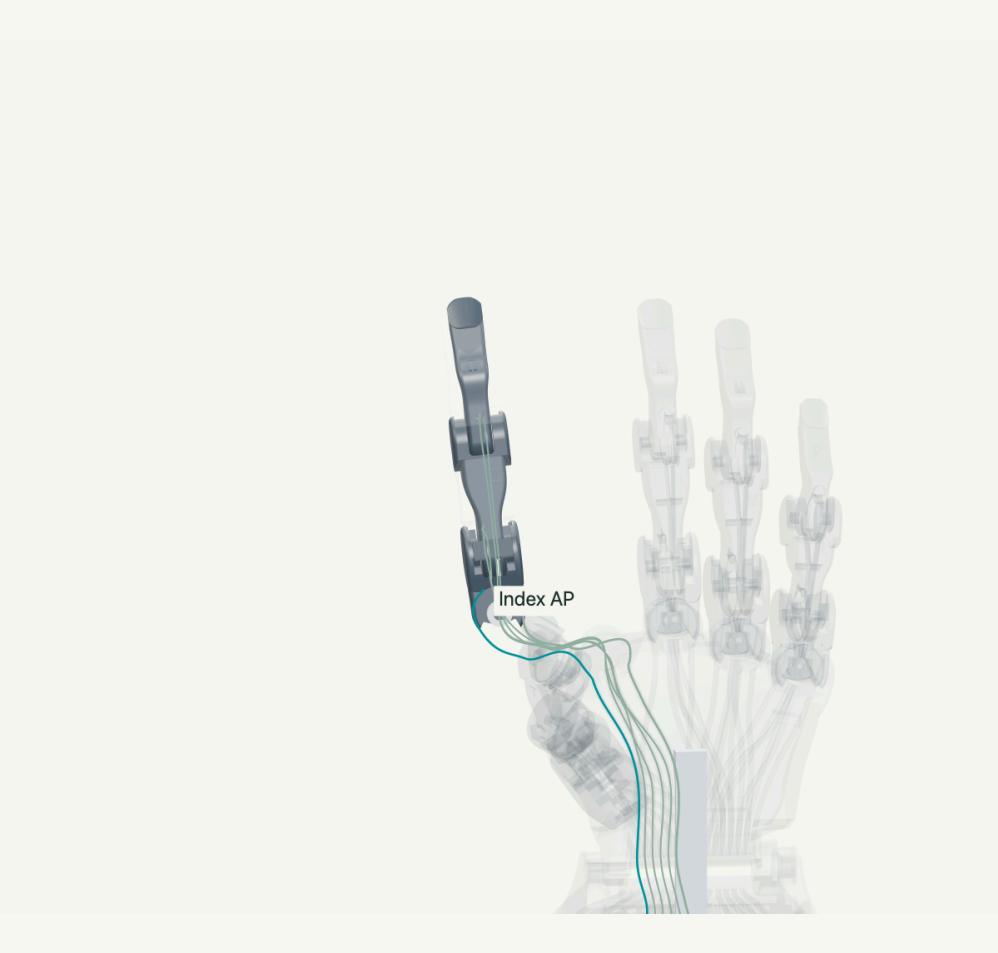


Continue through the wrist and into the tower.

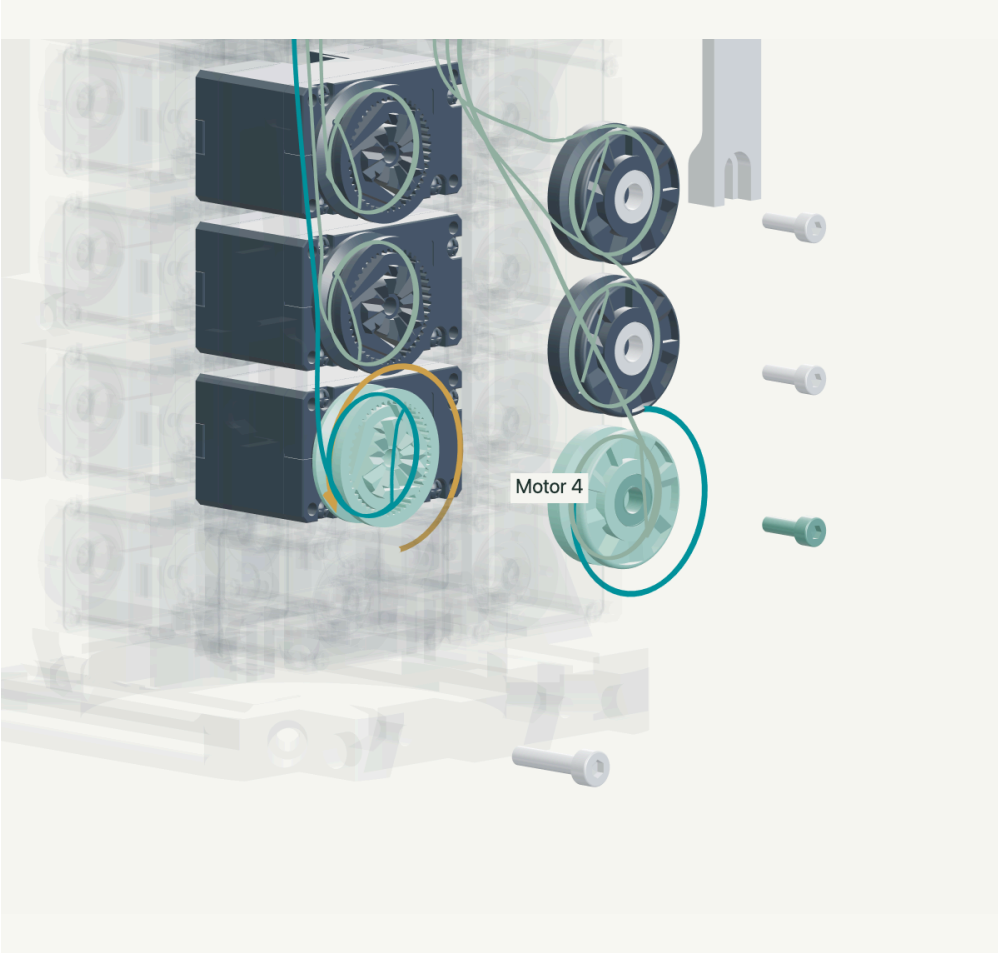
Follow the original route through the finger, carpals and wrist, then into the motor tower. Restore any displaced cords in order.

Use the original channels, with no crossed cords. Leave enough slack to refit the finger. Keep the original spool assignments and winding directions.

08 Pop in and re-spool



Pop the finger back into the hand.

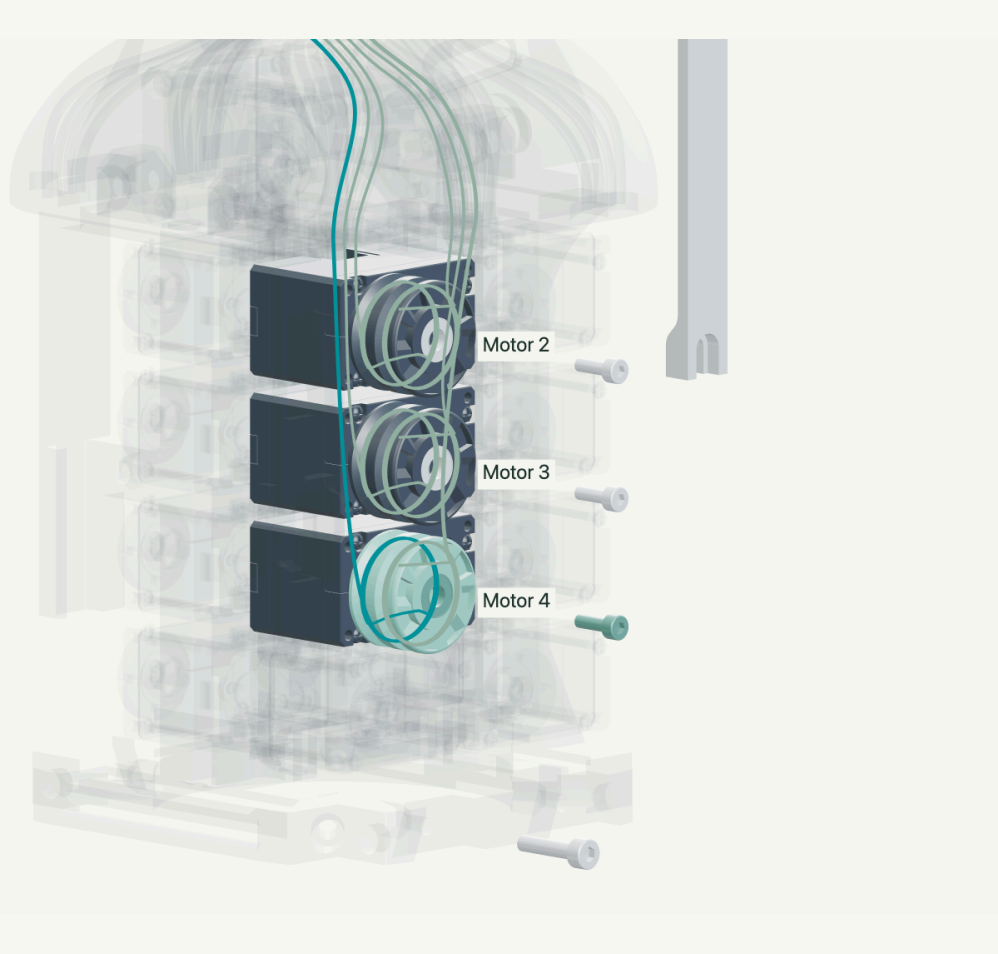


Rethread the freed ends; rewind the spools.

Pop the finger back in. Rethread and anchor the freed spool ends. Rewind those spools, then turn the others to take up their slack.

Use the matching assembly guide’s spool channels and Ashley stoppers. Wind three snug turns where cord length allows. Keep the recorded top/bottom assignments, with no cord crossed or caught behind a spool. Check each tendon with the others loose.

09 Refit, tension and close



Refit top spools and start the screws.



Tension, power off, tighten and refit the holder.

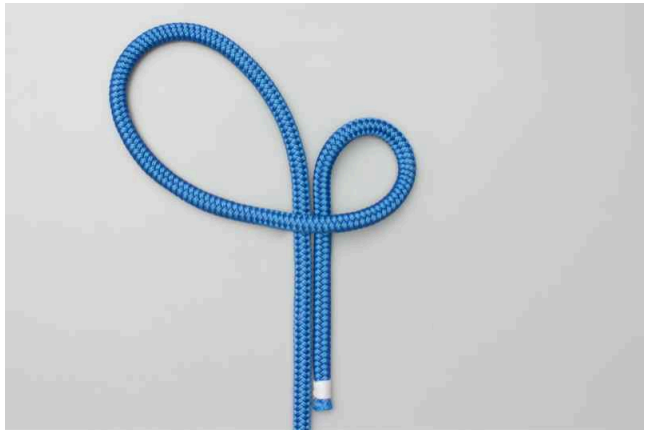
Refit the top spools and screws. Tension and check the finger, tighten the fixings, then refit and secure the spool holder.

Clear tools before powering on. Use Setup to repeat slack-test, tensioning and calibration for every joint loosened. Disconnect and switch board power off before the final mechanical tightening and holder refit. Test both directions; stop if a cord catches or moves the wrong joint.

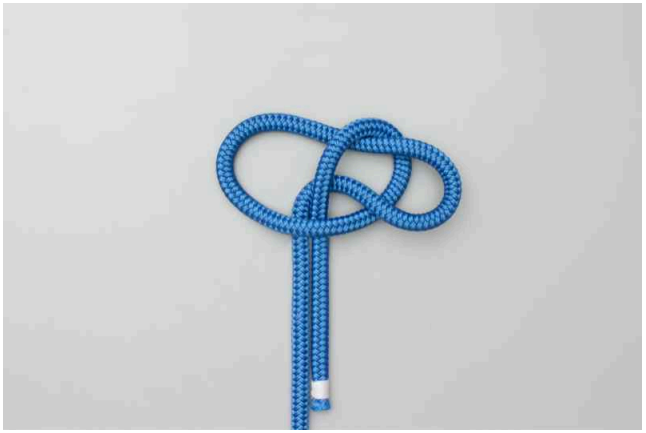
Ashley stopper · 1 / 2



01. Leave a working end long enough to handle. Fold the cord back to make a bight.



02. Fold the bight to form two loops. Keep track of which side leads to the long standing part.



03. Pass the loop on the standing-part side through the other loop.



04. Keep the new loop open. Bring the short working end toward it.



05. Feed the working end through the open loop.



06. Begin snugging the half knot while keeping the working end through the loop.

Ashley stopper · 2 / 2



07. Snug the half knot completely before tightening the rest.



08. Pull the short working end to remove its slack.



09. Pull the long standing part to finish tightening.



10. Dress the knot: arrange the turns neatly, then pull firmly to set it.



11. Inspect the underside. Look for three distinct lobes around the standing part.

Finish the AP anchor

Tie one single knot around the Ashley stopper, directly on top of it. Pull snug, then seat the anchor in the AP.

Spool anchors use the assembly guide’s Ashley stopper and three winding turns.